Homotopical Encryption: Hidden Hodge Operators and the Harmonic Cochain Problem

Chloe van der Vlugt

Abstract

We introduce the Harmonic Cochain Problem (HCP): recovering the operative harmonic component $h = P_{\ker(\Delta_1)}b$ from public data (α, b). The degree-one Hodge–Laplacian Δ_1 is induced by an operative hidden representative $x \in A_\alpha = \pi^{-1}(\alpha)$ of the public classifier value α. Its adjacent coboundaries δ_0, δ_1 determine the Hodge decomposition $b = h + e$, with $e \in \operatorname{im}(\delta_0) \oplus \operatorname{im}(\delta_1^*)$. We compare HCP with the closest-vector problem (CVP), distinguishing the field-linear image $\mathcal{V}_B(R) = \{Bx : x \in R^n\}$ from the discrete lattice $\Lambda_B = B\mathbb{Z}^n$. The former can be realized as a linear kernel. For an explicitly supplied inclusion $\Lambda_B \subseteq W$, orthogonal projection onto the harmonic subspace W preserves the closest lattice vectors. We then prove a hidden-operator separation: if publicly equivalent hidden operators disagree on harmonicity, a correct known-lattice CVP oracle cannot supply a uniformly exact public-data HCP solver.

We instantiate this non-identifiability with integral examples in a finite signed-incidence cellular model. Two presentations have the same public observation in a proved homological-test fragment of Helfer semantics but distinct nonzero harmonic targets. A weighted pair has the shared observation $b = (1, 2)$ with integer coefficients two and one, while each privately presented lattice has its own unique CVP answer. Further results separate full-sample recovery from an exact-component observation and reconstruct a subspace and its fixed-metric projector from full projective Plücker data. These are exact recovery and observation results, rather than asymptotic hardness or sampled encryption-security theorems. Protocol commitments and quantitative Reed–Solomon analysis are developed in separate companions.

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1 Introduction

1.1 Motivation and Contributions

Homotopical Encryption studies recovery from public homotopy-class data when the Hodge operator whose kernel is the message space is not part of the public input. In a known-lattice closest-vector problem, the lattice is presented before the search begins. In HCP, the recovery target is

$$h = P_H(b) \in \ker(\Delta_1),$$

but the public class does not include the degree-one Hodge–Laplacian Δ_1, its kernel, or the restriction maps from which Δ_1 is built. The private recovery object is therefore not a scalar secret or a vector in a published lattice; it is the harmonic cochain in the kernel of a hidden Hodge operator.

The operator dependence can be summarized as follows. A hidden representative determines the restriction data, which determine the adjacent coboundaries and hence the degree-one Hodge operator:

$$x \longmapsto \{R^x_{\sigma\to\tau}\}_{\sigma\prec\tau} \longmapsto (\delta_{0,x}, \delta_{1,x}) \longmapsto \Delta_{1,x} \longmapsto \big(\ker \Delta_{1,x}, P_{H_x}\big).$$

The recovery operation is Hodge-theoretic; "homotopical" describes the framework in which multiple hidden representatives may occupy the same public class while inducing different operators. The distinction is important: it is hidden-operator ambiguity, not whether a displayed matrix happens to have binary entries, that drives HCP.

This changes the comparison problem. A real linear span can be realized as the kernel of a suitable linear operator, but a discrete lattice cannot equal a nonzero real kernel. If an explicit lattice Λ is supplied inside a harmonic subspace $W = \ker(\Delta_W)$, orthogonal projection of the target onto W preserves the closest vectors of Λ. That bridge does not make public-data HCP a known-lattice problem: public homotopy-class data may be compatible with distinct hidden operators, distinct kernels, and different harmonicity judgments. The paper isolates this boundary and then separates it from the protocol observations and coding applications developed in the companion papers. The public classifier supplies the realized class value α. The full cochain $b = h + e$ is separate from the accepted exact commitment payload: its decoded exact cochain has zero harmonic projection in its specified Hodge system. A bound-observation relation ties the full sample to that payload, its digest, snapshot and lifecycle context. Releasing (α, b) requires an explicit disclosure or a specified extraction of that full sample; a proof about the exact payload alone does not expose it. The selected observation need not

present Δ_1 or $\ker(\Delta_1)$. If full Plücker coordinates of the operative kernel are additionally disclosed, however, they determine its fixed-metric projector and hence the HCP target from b. The principal notation is fixed globally as follows. Symbols used only inside a single proposition or construction are defined locally.

Symbol	Meaning
$p : E \to B$	Helfer-style fibration projection
$\mathcal{P}_X(x, y)$	path object from x to y in X
$\pi : A \to P$	path-respecting public classifier
$\alpha = \pi(x) \in P$	realized public classifier value
$A_\alpha = [x]_\pi$	fiber of hidden representatives with public label α
(α, b)	public HCP observation
$\Delta_A : A \to A \times A$	diagonal map in the fibrational semantics
$a \in C^0,\ c = \delta_0 a \in C^1$	degree-zero opening and committed exact one-cochain
$\delta_0, \delta_1, \Delta_1$	adjacent coboundaries and degree-one Hodge–Laplacian
$\ker(\Delta_1),\ P_H$	harmonic subspace and harmonic projection
$\Delta_W = I - P_W$	kernel-normalized projection operator associated with W

Table 1: Core notation used across the semantic and HCP layers.

This paper develops this boundary through the following statements.

(i) The Harmonic Cochain Problem (HCP): recovery of the operative component $h = P_H b$ for a hidden degree-one Hodge operator Δ_1.

(ii) A Hodge-structured noise model with cochain decomposition $b = h + e$, where $h \in \ker(\Delta_1)$ and $e \in \mathrm{im}(\delta_0) \oplus \mathrm{im}(\delta_1^*)$.

(iii) A CVP comparison separating field-linear image from discrete lattice: generated subspaces can be realized as kernels, and, when $\Lambda \subseteq W$ is explicitly supplied, projecting the target onto W preserves the discrete closest-vector condition on Λ.

(iv) A hidden-operator ambiguity theorem: publicly equivalent Hodge operators may disagree on harmonicity, so the CVP comparison does not make a correct explicit-lattice CVP oracle into a uniform public-data HCP solver.

(v) Constructed integral examples in the finite signed-incidence model: one homological public observation has distinct nonzero harmonic targets; a weighted pair shares $b = (1, 2)$ with integer coefficients two and one, with a unique CVP answer for each privately presented lattice.

(vi) Observation boundaries: exact-component extraction can lose full Hodge residual, whereas full projective data of the operative kernel determine its fixed-metric projector and harmonic target.

Finite Hodge theory follows the cellular-cochain framework of Hansen–Ghrist [HG19]; the algebraic model used here is specified in Section 2. Kernel realization and the Pythagorean projection identity are elementary linear-algebraic foundations. Helfer's homotopy-invariant semantics [Hel25] motivates the public classifier; the constructed examples prove only the explicitly named nonzero-cohomology-test fragment. These objects support the constructed observation ambiguities and recovery boundaries. The results establish neither an asymptotic CVP-hardness reduction nor a sampled encryption-security theorem; the semantic fragment and observation maps retain their explicit hypotheses.

The companion manuscript *Capacity-Radius Proximity for Smooth-Domain Reed–Solomon Codes* [vdV26a] develops the universal list-decoding and MCA results independently of HCP. The present paper supplies the HCP foundations and mathematical observation boundaries. Paper 2 [vdV26b] owns the concrete cochain commitment, public view and protocol evidence. Paper 3 owns the coding and quantitative projection analysis. Their necessary shared definitions are cited rather than repeated as new contributions; a source-to-code application still requires its own metric comparison.

The statement correspondence and formal proof boundaries are collected in the parity table after the conclusion (Section A).

2 The Harmonic Cochain Problem

The formal cellular model is algebraic: finite graded cell carriers, integer incidence numbers and face-to-coface linear stalk maps determine adjacent coboundaries satisfying $\delta_1\delta_0 = 0$. It adapts the finite Hodge setup of [HG19] without assuming a regular CW or simplicial realization for every admitted incidence structure. All cellular examples below use this declared model; stronger topological realization and the separate simplicial-coherence conditions require additional data.

Let A denote the type of hidden representatives, let P denote the type of public class data, and let

$$\pi : A \to P, \qquad \alpha \in \operatorname{im}(\pi), \qquad A_\alpha = \pi^{-1}(\alpha) = \{x \in A : \pi(x) = \alpha\}.$$

For $x \in A$, we also write $[x]_\pi := A_{\pi(x)}$. Thus, whenever $\alpha = \pi(x)$, the two descriptions agree: $A_\alpha = [x]_\pi$.

Definition 2.1 (Harmonic Cochain Problem)**.** *The* Harmonic Cochain Problem *(HCP) asks, given public data (α, b) consisting of a realized public classifier value $\alpha \in \mathrm{im}(\pi)$ and a cochain-facing observation b, to recover the operative harmonic component*

$$h = P_{\ker(\Delta_1)} b \in \ker(\Delta_1),$$

equivalently the harmonic term in the orthogonal Hodge decomposition of b. Here Δ_1 is the degree-one Hodge–Laplacian induced by the operative hidden representative $x \in A_\alpha = \pi^{-1}(\alpha)$ bound to the instance by its hidden witness.

In the Homotopical Encryption public-sample model, the visible cochain has the form

$$b = h + e, \qquad h \in \ker(\Delta_1), \qquad e \in \mathrm{im}(\delta_0) \oplus \mathrm{im}(\delta_1^*).$$

Here h is the degree-one harmonic cochain, equivalently a harmonic representative of its degree-one cohomology class, and e is Hodge-structured noise. The hidden-operator ambiguity studied below is the condition that the public data do not determine a unique Δ_1 or $\ker(\Delta_1)$. The algebraic recovery chain is

$$x \longmapsto \{R_{\sigma\to\tau}\}_{\sigma\prec\tau} \longmapsto (\delta_0, \delta_1) \longmapsto \Delta_1 \longmapsto P_H : C^1 \to \ker(\Delta_1),$$

where $x \in A_\alpha$ is a hidden representative and $R_{\sigma\to\tau}$ are the face-to-coface restriction maps, including vertex-to-edge and edge-to-face maps, that determine the adjacent coboundaries

$$\delta_0 : C^0 \to C^1, \qquad \delta_1 : C^1 \to C^2,$$

and

$$\Delta_1 = \delta_0\delta_0^* + \delta_1^*\delta_1$$

is the induced degree-one Hodge–Laplacian. The map P_H is the orthogonal projection onto $\ker(\Delta_1)$. When the degree is fixed, we write Δ for this operative Laplacian. A concrete public artifact may commit to or certify values derived from this chain without revealing the restriction maps or the induced operator.

Definition 2.2 (Matrix harmonicity)**.** *For a hidden linear Hodge operator Δ, a cochain $h \in \mathbb{R}^N$ is Δ-harmonic when*

$$\Delta h = 0.$$

Thus the harmonic kernel is $\ker(\Delta)$*. In the projection model this agrees with the condition $h \in W$ for the kernel-normalized operator $\Delta_W = I - P_W$.*

Definition 2.3 (Kernel-normalized operator, harmonic projection, and residual)**.** *Let $V = \mathbb{R}^n$ be equipped with its standard inner product, and let $W \subset V$ be a linear subspace. Write*

$$P_W : V \to W \subset V \qquad \text{and} \qquad P_{W^\perp} : V \to W^\perp \subset V$$

for the orthogonal projections. The kernel-normalized projection operator attached to W is

$$\Delta_W = I - P_W.$$

For $b \in V$, define the residual vector and residual score by

$$r_W(b) = \Delta_W b, \qquad \rho_W(b) = \|r_W(b)\|.$$

We say that b is W-harmonic when $r_W(b) = 0$. This operator records exactly the harmonic kernel and its orthogonal projection. It is not asserted to equal a cellular Hodge–Laplacian $\delta_0\delta_0^* + \delta_1^*\delta_1$, or to have the same positive spectrum, unless the cellular Laplacian additionally acts as the identity on $W^\perp$.

Proposition 2.4 (Projection harmonicity equivalences)**.** *For every $b \in V$,*

$$r_W(b) = P_{W^\perp} b, \qquad \rho_W(b) = \|P_{W^\perp} b\|, \qquad \rho_W(b) \geq 0.$$

Moreover

$$P_W b \in W, \qquad r_W(P_W b) = 0,$$

and the following conditions are equivalent:

$$r_W(b) = 0 \iff \rho_W(b) = 0 \iff b \in W.$$

Proof. Since $V = W \oplus W^\perp$ orthogonally, every $b \in V$ decomposes uniquely as

$$b = P_W b + P_{W^\perp} b.$$

Hence

$$(I - P_W)b = b - P_W b = P_{W^\perp} b,$$

giving $r_W(b) = P_{W^\perp} b$ and the displayed formula for $\rho_W(b)$. Nonnegativity is the nonnegativity of the norm. The projection $P_W b$ lies in W, so its orthogonal component is zero; therefore $r_W(P_W b) = 0$. Finally, $r_W(b) = 0$ iff $P_{W^\perp} b = 0$, which holds iff $b \in W$. The equivalence with $\rho_W(b) = 0$ follows from $\|x\| = 0$ iff $x = 0$. □

Proposition 2.5 (Hodge decomposition surface)**.** *Let the three consecutive cochain spaces be finite-dimensional real inner product spaces, and let the displayed stars denote adjoints. Suppose*

$$d_{\text{next}} d_{\text{prev}} = 0, \qquad \Delta = d_{\text{prev}} d_{\text{prev}}^* + d_{\text{next}}^* d_{\text{next}}.$$

Then every cochain b admits a decomposition

$$b = h + e_1 + e_2$$

with

$$\Delta h = 0, \qquad e_1 \in \operatorname{im}(d_{\text{prev}}), \qquad e_2 \in \operatorname{im}(d_{\text{next}}^*),$$

and

$$\langle h, e_1 \rangle = \langle h, e_2 \rangle = \langle e_1, e_2 \rangle = 0.$$

The decomposition is unique.

Proof. Each summand defining Δ is self-adjoint, so Δ is self-adjoint. In finite dimension, $\operatorname{im}(\Delta) = (\ker \Delta)^\perp$, and hence the ambient cochain space decomposes as

$$\ker(\Delta) \oplus \operatorname{im}(\Delta).$$

Since

$$\operatorname{im}(\Delta) \subseteq \operatorname{im}(d_{\text{prev}}) + \operatorname{im}(d_{\text{next}}^*),$$

every cochain b can be written as $h + e_1 + e_2$ with $h \in \ker \Delta$, $e_1 \in \operatorname{im}(d_{\text{prev}})$, and $e_2 \in \operatorname{im}(d_{\text{next}}^*)$. From $\Delta h = 0$, one obtains

$$\|d_{\text{prev}}^* h\|^2 + \|d_{\text{next}} h\|^2 = 0,$$

hence $d_{\text{prev}}^* h = 0$ and $d_{\text{next}} h = 0$. These identities give $\langle h, e_1 \rangle = 0$ and $\langle h, e_2 \rangle = 0$, while $d_{\text{next}} d_{\text{prev}} = 0$ gives $\langle e_1, e_2 \rangle = 0$. Hence the harmonic, exact, and coexact subspaces are pairwise orthogonal. Their sum is therefore direct, which makes the three components unique. □

2.1 Hodge Encryption and Correctness

For a hidden representative x, write

$$\mathcal{H}_x = \ker(\Delta_x), \qquad \mathcal{E}_x = \mathcal{H}_x^{\perp}.$$

The Hodge decomposition identifies the noise space as the orthogonal direct sum

$$\mathcal{E}_x = \operatorname{im}(\delta_{0,x}) \oplus^{\perp} \operatorname{im}(\delta_{1,x}^{*}),$$

and gives the orthogonal decomposition

$$C_x^1 = \mathcal{H}_x \oplus^{\perp} \mathcal{E}_x.$$

Consequently,

$$\begin{aligned} \operatorname{combine}_x : \mathcal{H}_x \times \mathcal{E}_x &\longrightarrow C_x^1, & (h, e) &\longmapsto h + e, \\ \operatorname{split}_x : C_x^1 &\longrightarrow \mathcal{H}_x \times \mathcal{E}_x, & b &\longmapsto \big(P_{\mathcal{H}_x} b, P_{\mathcal{E}_x} b\big) \end{aligned}$$

are inverse linear bijections. Orthogonality also gives

$$\|h + e\|^2 = \|h\|^2 + \|e\|^2.$$

This bijection is the algebraic encryption/decryption construction. Given a harmonic message $h \in \mathcal{H}_x$, randomized Hodge encryption samples, or otherwise chooses, admissible noise $e \in \mathcal{E}_x$ and sets

$$\mathsf{Enc}_x(h; e) = \operatorname{combine}_x(h, e) = h + e.$$

Full coordinate recovery is split_x; message decryption retains its harmonic coordinate:

$$\mathsf{Dec}_x(b) = \operatorname{pr}_{\mathcal{H}_x}\big(\operatorname{split}_x(b)\big) = P_{\mathcal{H}_x} b.$$

For every admissible noise value,

$$\mathsf{Dec}_x\big(\mathsf{Enc}_x(h; e)\big) = h, \qquad \operatorname{split}_x\big(\operatorname{combine}_x(h, e)\big) = (h, e).$$

Thus the bijection is between the full message–randomness pair and C_x^1. After randomness is sampled, Enc_x is a randomized procedure and Dec_x is its message-correct left inverse; this does not assert that a randomized map from messages alone is a bijection.

The subscript x is essential. The hidden representative—and, in a keyed instantiation, the private material selecting or opening it—determines Δ_x, the two summands, and the splitting map. The correctness identity is pointwise

and requires no uniform distribution on an infinite noise space. It is not an indistinguishability or one-wayness claim; those require separate assumptions about key generation, randomness, and the public view.

When several presentation-dependent spaces C_x^1 are compared using one public coordinate vector, we fix isometric coordinate charts $\iota_x : C_x^1 \xrightarrow{\sim} V$ into a common finite-dimensional real inner-product space V. Operators denoted $\Delta_x \in \mathrm{End}(V)$ below are the corresponding conjugated operators. The charts are comparison data; they are not asserted to be recoverable from the public classifier.

Proposition 2.6 (Exact-component obstruction)**.** *Suppose $v \in C^2$ and $\delta_1^* v \neq 0$. On the unrestricted Hodge-noise carrier, the samples*

$$b_t = t\delta_1^* v, \qquad t \in \mathbb{R},$$

have zero harmonic message and zero exact component, while

$$\|P_{\mathcal{H}^\perp} b_t\| = |t| \, \|\delta_1^* v\|.$$

Thus this full-observation residual does not factor through the exact component. If an extraction depends only on that component and its zero image has target distance zero, it cannot preserve any positive target far radius uniformly on this unrestricted carrier, at any source radius.

Proof. A coexact vector is orthogonal to $\mathcal{H}$, so its orthogonal projection onto $\mathcal{H}^\perp$ is itself. Homogeneity gives the displayed norm. The exact-component view is constantly zero, whereas this norm is zero at $t = 0$ and positive at $t = 1$, proving nonfactorization. For any fixed source radius, choose positive t large enough to exceed it. The extracted value still has distance zero, contradicting every positive target far radius. □

This statement concerns a map factoring through the decoded exact component. Equal decoded components do not by themselves imply equal raw payload bytes for an arbitrary decoder. A restricted noise or authenticated realization profile may exclude this family; a theorem about such a profile requires its explicit hypotheses. The statement does not identify distance to exactness with the full HCP residual.

2.2 Public and Hidden Data

Definition 2.7 (Public observation)**.** *A public observation is the data exposed to a verifier or adversary without revealing a hidden representative. In the*

abstract model it consists of a realized public classifier value α, together with a cochain-facing observation b. In the Homotopical Encryption instantiation, the fully public handles are hashes, roots, topology hashes, nullifiers, and bounded public summaries; cochain commitments and proof-carrying cochain payloads sit in the verifier-facing artifact layer; full cochains and openings remain witness data unless an explicit protocol path exposes them.

Thus (α, b) is the protocol-independent vector-level abstraction of the artifact-and-commitment layer developed in Paper 2 [vdV26b]: α records the public classifier value, while b is the full cochain actually released by the selected observation profile. Acceptance of an exact payload, an opening of its exact component, or a proof about that component does not by itself expose b. Instantiating the vector-level interface requires a relation binding the full sample to the artifact and an explicit disclosure or specified extraction of that sample. Section 5 records the mathematical distinction; Paper 2 supplies its concrete bound-observation contracts. The hidden-kernel ambiguity comparison uses the resulting mathematical pair; equality of a larger artifact or storage view requires its own hypotheses.

The public observation is not the hidden Hodge system. It classifies a public equivalence class of possible hidden representatives. Fibrational and homotopical semantics enter through this distinction: public equality is equality after applying a public classifier, not strict equality of hidden representatives. Section 4 develops this semantic layer.

Definition 2.8 (Hidden-operator public projection)**.** *For a public class type P, a public projection is a map*

$$\pi_\Delta : \{\textit{hidden Hodge operators } \Delta\} \longrightarrow P.$$

Two hidden operators are publicly equivalent when $\pi_\Delta(\Delta) = \pi_\Delta(\Delta')$. The hidden operator fiber over $\alpha \in P$ is

$$\mathrm{Fib}_{\pi_\Delta}(\alpha) = \pi_\Delta^{-1}(\alpha) = \{\Delta : \pi_\Delta(\Delta) = \alpha\},$$

and the hidden class of Δ is $\mathrm{Fib}_{\pi_\Delta}(\pi_\Delta(\Delta))$.

Definition 2.9 (Lossiness and kernel ambiguity)**.** *The public projection π_Δ is* lossy *if there exist $\Delta \neq \Delta'$ with $\pi_\Delta(\Delta) = \pi_\Delta(\Delta')$. The two operators have the same harmonic kernel if*

$$\forall v, \quad \Delta v = 0 \Longleftrightarrow \Delta' v = 0.$$

The projection is kernel-ambiguous *if there are publicly equivalent operators Δ, Δ' whose harmonic kernels are not the same.*

Definition 2.10 (Public-compatible Hodge ambiguity)**.** *Public-compatible Hodge ambiguity means that there exist publicly equivalent operators Δ, Δ' and a cochain-facing observation b such that*

$$\Delta b = 0 \qquad \textit{but} \qquad \Delta' b \neq 0.$$

Definition 2.11 (Typed HCP semantic realization)**.** *Fix a multi-sorted algebraic signature σ, a model category $\mathcal{C}$ with terminal object and binary products, and, for each hidden representative $x \in A$, a base interpretation $M_x : \sigma \to \mathcal{C}_{\mathrm{f}}$ together with homotopical formula semantics over M_x. Write*

$$\mathrm{Th}(M_x) = \{\varphi \textit{ closed} : M_x \vDash \varphi\}$$

for its closed first-order theory. Use the chosen coordinate charts to compare cochains in a common finite-dimensional real inner-product space V. A typed HCP semantic realization consists additionally of maps

$$\begin{aligned} x &\longmapsto \Delta_x \in \mathrm{End}(V), \\ \kappa &: \{\textit{closed theories}\} \longrightarrow P, \\ \pi_\Delta &: \{\textit{Hodge operators}\} \longrightarrow P, \end{aligned}$$

satisfying the two factorization identities

$$\pi(x) = \kappa(\mathrm{Th}(M_x)), \qquad \pi_\Delta(\Delta_x) = \pi(x).$$

The Laplacian Δ_x is presentation-level Hodge data extending the semantic core M_x; it is not silently included among the symbols whose closed theory is asserted to be homotopy invariant.

Proposition 2.12 (Helfer-to-HCP classifier bridge)**.** *Let $x, y \in A$ belong to a typed HCP semantic realization. If M_x and M_y are homotopy equivalent interpretations in the sense of Helfer, then*

$$\pi(x) = \pi(y) \qquad \textit{and} \qquad \pi_\Delta(\Delta_x) = \pi_\Delta(\Delta_y).$$

If, independently, there is a cochain b for which

$$\Delta_x b = 0, \qquad \Delta_y b \neq 0,$$

then π_Δ has public-compatible Hodge ambiguity, and hence is kernel-ambiguous and lossy.

Proof. The closed-formula case of Helfer's homotopy-invariance theorem [Hel25, Theorem 12.3] gives, for every closed σ-formula φ,

$$M_x \models \varphi \quad \Longleftrightarrow \quad M_y \models \varphi.$$

Hence $\mathrm{Th}(M_x) = \mathrm{Th}(M_y)$. Applying κ and then the two factorization identities gives

$$\pi(x) = \pi(y), \qquad \pi_\Delta(\Delta_x) = \pi_\Delta(\Delta_y).$$

The additional displayed harmonicity and non-harmonicity conditions now exhibit (Δ_x, Δ_y, b) as a public-compatible Hodge-ambiguity witness. The ambiguity hierarchy below supplies kernel ambiguity and lossiness. □

The direction in Proposition 2.12 is one-way. Homotopy-equivalent interpretations have the same factored public value, but equality of public values need not reconstruct a homotopy equivalence: a classifier fiber may be strictly coarser than an equivalence class. Likewise, the equivalence itself does not produce kernel disagreement. If Δ-harmonicity were included in the invariant first-order signature, and the common coordinate b were named compatibly in both interpretations, Helfer's theorem would instead rule out the displayed disagreement.

There is a second, geometric distinction. Suppose an equivalence of hidden presentations induces degreewise linear isometric equivalences U_0, U_1, U_2 satisfying

$$U_1 \delta_{0,x} = \delta_{0,y} U_0, \qquad U_2 \delta_{1,x} = \delta_{1,y} U_1.$$

After conjugating by the fixed coordinate charts, U_1 is an isometry of the common comparison space V. Because the U_k preserve inner products, the adjoints transport as well, and hence

$$\Delta_y U_1 = U_1 \Delta_x, \qquad U_1(\ker \Delta_x) = \ker \Delta_y.$$

The corresponding orthogonal projections commute with transport:

$$U_1 P_{\ker \Delta_x} b = P_{\ker \Delta_y} U_1 b.$$

Thus a transported observation preserves both the operative target and harmonicity:

$$\Delta_x b = 0 \quad \Longleftrightarrow \quad \Delta_y(U_1 b) = 0.$$

The ambiguity witness in Proposition 2.12 deliberately compares the same untransported public coordinate $b \in V$, not the pair $(b, U_1 b)$. Under the displayed conjugacy, its extra separation condition is equivalently

$$b \in \ker \Delta_x, \qquad b \notin U_1(\ker \Delta_x).$$

It therefore requires a nonstabilization witness in addition to the equivalence, and it models a public view that neither reveals nor applies the private transport U_1. If the observation were transported together with the hidden presentation, this particular ambiguity would disappear.

Proposition 2.13 (Ambiguity hierarchy)**.** *Public-compatible Hodge ambiguity implies kernel ambiguity, and kernel ambiguity implies that the public projection is lossy.*

Proof. If b is harmonic for Δ but not for Δ', then the two operators cannot have the same harmonic kernel. This gives kernel ambiguity. If publicly equivalent operators have different kernels, they cannot be the same operator, since equality of operators implies equality of kernels. Therefore the public projection identifies two distinct hidden operators. □

Definition 2.14 (Hidden Hodge witness)**.** *A hidden Hodge witness for a public observation (α, b) is a hidden representative $x \in A$ satisfying $\pi(x) = \alpha$. Through the typed realization, it selects the operative operator Δ_x. The witness and b then determine, rather than independently assume, the coordinates*

$$h_x(b) := P_{\ker(\Delta_x)} b, \qquad e_x(b) := b - h_x(b).$$

For a genuine degree-one Hodge operator $\Delta_x = \delta_{0,x}\delta_{0,x}^* + \delta_{1,x}^*\delta_{1,x}$, the Hodge decomposition gives

$$h_x(b) \in \ker(\Delta_x), \qquad e_x(b) \in \operatorname{im}(\delta_{0,x}) \oplus \operatorname{im}(\delta_{1,x}^*), \qquad b = h_x(b) + e_x(b).$$

The hidden data include the fibration representative, restriction maps, the induced Laplacian, its harmonic kernel, and private witness/cochain data. Public data may bind to this hidden data through commitments or proof-carrying artifacts, but it does not present the operative kernel.

Definition 2.15 (HCP solution)**.** *Relative to an operative hidden witness x, a solution to an HCP instance (α, b) is the cochain $h' = h_x(b) = P_{\ker(\Delta_x)} b$. A public-data-only solver is* uniformly exact *if the same output has this value for every hidden witness compatible with its public input. The hidden-operator separation concerns this uniform contract; it does not deny that decryption succeeds once the operative private geometry is supplied.*

2.3 Recovery Target

The recovery target is the orthogonal projection of b into the operative kernel, not the entire kernel subspace. Thus CVP asks for a closest vector in a

presented lattice, while HCP asks for $P_{\ker(\Delta_x)}b$ for the operative hidden representative.

In the metric variant used for comparison with CVP, the continuous target is the closest harmonic cochain to b. Presenting Δ or a basis for $\ker(\Delta)$ presents a real subspace, not a discrete lattice. A genuine CVP instance additionally requires an explicit lattice $\Lambda \subseteq \ker(\Delta)$; Proposition 3.2 shows that projection onto the kernel preserves the closest vectors of that supplied lattice. The hidden-operator model withholds the operative subspace itself.

2.4 Computational Model

The computational model separates three layers.

(i) The mathematical layer defines public projections, hidden operator fibers, harmonicity, and kernel ambiguity.

(ii) The external known-lattice CVP hardness interface records a computational assumption, separate from the proved exact-recovery obstruction; no asymptotic hardness reduction is asserted.

(iii) Runtime requirements such as byte parsing, canonical encodings, hash domain separation, proof verification, and state transition are outside the mathematical HCP definition.

This paper proves the mathematical comparisons and states the external CVP interface for context. Paper 2 owns concrete protocol and runtime claims; Paper 3 owns coding and proximity analysis.

3 CVP Comparison and Kernel Ambiguity

CVP asks for a closest vector in an explicitly presented lattice [MG02]. HCP asks for the projection of b into the kernel of the operative hidden Hodge operator. In the Homotopical Encryption model, public data need not determine the corresponding harmonic kernel.

3.1 Known-Lattice CVP

An explicit Euclidean CVP instance consists of a full-column-rank generator matrix $B \in \mathbb{R}^{m\times n}$ and a target $t \in \mathbb{R}^m$. Its discrete lattice is

$$\Lambda_B = B\mathbb{Z}^n = \{Bz : z \in \mathbb{Z}^n\}.$$

The injectivity of $z \mapsto Bz$ records that the displayed columns are a lattice basis. A correct closest-vector answer is a vector $v \in \Lambda_B$ such that

$$\|t - v\|^2 \leq \|t - w\|^2 \qquad \text{for every } w \in \Lambda_B.$$

Thus, for fixed dimensions, a CVP oracle O is correct when

$$\begin{aligned} \mathsf{CVP}_{\mathrm{corr}}(O) \quad \Longleftrightarrow \quad & \forall \text{ full-column-rank } B, t, \\ & O(B,t) \in \Lambda_B \quad \text{and} \quad \forall w \in \Lambda_B, \\ & \|t - O(B,t)\|^2 \leq \|t - w\|^2. \end{aligned}$$

The weaker output condition is merely lattice membership:

$$\forall B, t,\ O(B,t) \in \Lambda_B.$$

Fix a class $\mathcal{C}$ of admissible algorithms, such as polynomial-time or probabilistic polynomial-time algorithms. The external CVP hardness assumption is the schema

$$\mathsf{CVP}_{\mathrm{hard}}(\mathcal{C}) \quad \Longleftrightarrow \quad \neg\exists O,\ O \in \mathcal{C} \wedge \mathsf{CVP}_{\mathrm{corr}}(O).$$

This is the separation between the known-lattice interface and the hidden-operator interface: a CVP oracle receives B and t, while a public HCP procedure sees only public class values, hashes, commitments, and verifier-facing artifacts unless extra hidden-operator information is supplied.

3.2 Presented Linear Images

Let R be a field. The field-linear image of B is a different object:

$$\mathcal{V}_B(R) = BR^n = \{Bx : x \in R^n\}.$$

Over a field this is a linear subspace, not the discrete lattice $B\mathbb{Z}^n$. The following comparison is therefore linear algebra; it is not a reduction from discrete CVP.

Let B present $\mathcal{V}_B(R)$. If a matrix Δ satisfies $\ker(\Delta) = \mathcal{V}_B(R)$, then any solver satisfying

$$\Delta S(\Delta, t) = 0$$

returns $S(\Delta, t) \in \mathcal{V}_B(R)$.

Proposition 3.1 (Kernel realization for explicit generated subspaces)**.** *Let $B \in R^{n\times n}$ be an explicit square generator matrix over a field and set $\mathcal{V}_B(R) = \{Bx : x \in R^n\}$. There exists a matrix Δ such that*

$$\ker(\Delta) = \mathcal{V}_B(R).$$

Consequently, any solver $S(\Delta, b)$ whose output is Δ-harmonic for every presented Δ returns a vector in the explicit generated subspace $\mathcal{V}_B(R)$ when run on this realized operator.

Proof. Let $B \in R^{n\times n}$ be an explicit generator matrix over a field R, set

$$\mathcal{V}_B(R) = \{Bx : x \in R^n\} \subseteq R^n,$$

and let $t \in R^n$ be the solver input. We construct a matrix Δ with

$$\ker(\Delta) = \mathcal{V}_B(R).$$

Put $V = \mathcal{V}_B(R)$. Since R^n is finite-dimensional over a field, choose a complementary subspace W with

$$V + W = R^n, \qquad V \cap W = 0.$$

Thus every $y \in R^n$ has a unique decomposition

$$y = v + w, \qquad v \in V, \quad w \in W.$$

Equivalently, there is a unique element $w_y \in W$ such that $y - w_y \in V$. Define

$$P(y) = w_y.$$

The uniqueness of the $V \oplus W$ decomposition gives linearity: for $y, z \in R^n$, the vector $P(y) + P(z)$ lies in W, and

$$(y + z) - (P(y) + P(z)) = (y - P(y)) + (z - P(z)) \in V,$$

so $P(y + z) = P(y) + P(z)$. Similarly, for $a \in R$, $aP(y) \in W$ and

$$ay - aP(y) = a(y - P(y)) \in V,$$

so $P(ay) = aP(y)$.

We now compute the kernel. If $y \in V$, then $0 \in W$ and $y - 0 \in V$, hence uniqueness gives $P(y) = 0$. Conversely, if $P(y) = 0$, then $y - P(y) = y \in V$. Therefore

$$\ker P = V = \mathcal{V}_B(R).$$

Let Δ be the matrix of the linear map P. Then, for every $y \in R^n$,

$$\Delta y = 0 \quad \Longleftrightarrow \quad y \in \ker P \quad \Longleftrightarrow \quad y \in \mathcal{V}_B(R) \quad \Longleftrightarrow \quad \exists x,\ Bx = y.$$

Let $S : R^{n\times n} \times R^n \to R^n$ be a presented-operator Hodge solver satisfying

$$\forall \Delta\, b, \qquad \Delta(S(\Delta, b)) = 0.$$

Applying this to the Δ constructed above and $b = t$ gives

$$\Delta(S(\Delta, t)) = 0.$$

By the kernel calculation, there exists x with

$$Bx = S(\Delta, t).$$

Hence $S(\Delta, t) \in \mathcal{V}_B(R)$. □

When $R = \mathbb{R}$, if a presented-operator solver additionally minimizes Euclidean distance among Δ-harmonic vectors, the same kernel equality says that it returns the continuous closest point of $\mathcal{V}_B(\mathbb{R})$. This is still not discrete CVP unless a separate lattice is supplied.

3.3 Explicit Harmonic-Lattice Bridge

The discrete bridge requires both kinds of data: a real harmonic subspace and an integer lattice contained in it. Let $W \subseteq \mathbb{R}^m$ be the operative harmonic subspace, let P_W be its orthogonal projection, and let $\Lambda_B = B\mathbb{Z}^n \subseteq W$ be an explicitly presented lattice. Set

$$\Delta_W = I - P_W.$$

The inclusion $\Lambda_B \subseteq W$ does not identify the discrete lattice with the real subspace.

Proposition 3.2 (Discrete CVP through harmonic projection)**.** *For $t \in \mathbb{R}^m$, put $t_W = P_W t$. Then:*

(i) t_W and every $\lambda \in \Lambda_B$ are Δ_W-harmonic;

(ii) for every $\lambda \in \Lambda_B$,

$$\|t - \lambda\|^2 = \|t - t_W\|^2 + \|t_W - \lambda\|^2;$$

(iii) a lattice vector λ is closest to t in Λ_B if and only if it is closest to t_W in Λ_B.

Proof. Since $t_W \in W$, every $\lambda \in \Lambda_B$ also lies in W, and $\ker(\Delta_W) = W$, the harmonicity claims follow. For the distance identity, write

$$t - \lambda = (t - t_W) + (t_W - \lambda).$$

Orthogonal projection gives $t - t_W \in W^\perp$, while $t_W - \lambda \in W$. The two summands are orthogonal, so the Pythagorean identity gives (ii). The term $\|t - t_W\|^2$ is independent of λ; consequently, for any $\lambda, \mu \in \Lambda_B$,

$$\|t - \lambda\|^2 \leq \|t - \mu\|^2 \quad \iff \quad \|t_W - \lambda\|^2 \leq \|t_W - \mu\|^2.$$

This is exactly (iii). □

The continuous projection $t_W = P_W t$ is the closest harmonic vector to t, but it need not lie in Λ_B. Thus projection transports the target to the harmonic subspace while preserving the discrete minimizers; it does not perform the remaining lattice search. The proposition also presumes that W and the inclusion $\Lambda_B \subseteq W$ have been supplied. It does not discover the operative hidden subspace from public class data.

3.4 Oracle Insufficiency

The oracle-insufficiency direction uses the typed realization from Definition 2.11. A public Hodge solver may call an explicit-lattice CVP oracle and receives the actual public observation (α, b), but it receives no hidden representative or operative Laplacian. Correctness is required only for representatives admitted by the realization, rather than for every matrix of the ambient size.

Definition 3.3 (Public-data Hodge solver contract)**.** *Let*

$$S : \mathcal{O}_{\mathrm{CVP}} \times (P \times \mathbb{R}^n) \longrightarrow \mathbb{R}^n,$$

where $\mathcal{O}_{\mathrm{CVP}}$ is the class of explicit-lattice CVP oracles. For a realization $x \mapsto (M_x, \Delta_x)$ and $O \in \mathcal{O}_{\mathrm{CVP}}$, define

$$\begin{aligned} \mathsf{H}_O(S) &\iff \forall x : A, \quad \forall b \in \mathbb{R}^n, \quad \Delta_x\big(S(O, (\pi(x), b))\big) = 0, \\ \mathsf{F}_O(S) &\iff \forall x : A, \quad \forall b \in \mathbb{R}^n, \quad \Delta_x b = 0 \Rightarrow S(O, (\pi(x), b)) = b. \end{aligned}$$

The public-data-only contract is

$$\exists S, \quad \forall O, \quad \mathsf{CVP}_{\mathrm{corr}}(O) \Rightarrow \big(\mathsf{H}_O(S) \wedge \mathsf{F}_O(S)\big).$$

Proposition 3.4 (Typed HCP ambiguity and CVP oracle insufficiency)**.** *Let $x, y \in A$ be representatives in a typed HCP semantic realization. Suppose that M_x and M_y are homotopy equivalent and that some $b \in \mathbb{R}^n$ satisfies*

$$\Delta_x b = 0, \qquad \Delta_y b \neq 0.$$

If at least one correct explicit-lattice CVP oracle exists, then

$$\neg\exists S, \quad \forall O, \quad \mathsf{CVP}_{\mathrm{corr}}(O) \Rightarrow \big(\mathsf{H}_O(S) \wedge \mathsf{F}_O(S)\big).$$

Hence a correct oracle for explicit known-lattice CVP does not by itself determine the operative Hodge–Laplacian kernel.

Proof. By Proposition 2.12, $\pi(x) = \pi(y)$. Suppose $\exists O, \mathsf{CVP}_{\mathrm{corr}}(O)$, and choose such an O. Suppose, for contradiction, that S satisfies the displayed public-data-only contract. Then

$$\mathsf{H}_O(S) \wedge \mathsf{F}_O(S).$$

Since $\Delta_x b = 0$, exactness at x gives

$$S(O, (\pi(x), b)) = b.$$

Soundness at y gives

$$\Delta_y S(O, (\pi(y), b)) = 0.$$

Replacing $\pi(y)$ by $\pi(x)$ and using the exactness equality yields $\Delta_y b = 0$, contradicting the separation hypothesis. □

The obstruction is kernel ambiguity: the same public observation may be compatible with multiple hidden operators whose harmonic kernels differ. A CVP oracle can solve a closest-vector problem after the lattice is presented. The bridge in Proposition 3.2 applies only after the operative harmonic subspace and the lattice inclusion have also been supplied; it does not determine which hidden kernel is compatible with the public data.

For every dimension at least two, consider the elementary matrix witness $L = 0$, $L' = \mathrm{diag}(1, 0, \ldots, 0)$, and $b = e_1$ under the constant projection to a one-point public type. This is a sanity example showing that the abstract ambiguity predicates are inhabited. It is not used as evidence that an arbitrary HCP realization admits the required semantic and Hodge extension witness. That witness remains an explicit hypothesis of Proposition 3.4; the following family constructs one for its stated semantic fragment.

3.5 A Constructed Integral Cellular Family

The preceding implication has a concrete instantiation in a stated homological fragment of the typed Helfer interface. Take one vertex, one edge and one face with incidence numbers $\partial_1 = 0$, $\partial_2 = 1$, and stalks $0, \mathbb{R}^2, \mathbb{R}$, respectively. This is the algebraic finite signed-incidence model of Section 2; no regular CW or simplicial realization of this displayed base is asserted. Two cellular sheaves in that model have

$$\delta_{0,0} = \delta_{0,1} = 0, \qquad \delta_{1,0}(x_0, x_1) = x_0, \qquad \delta_{1,1}(x_0, x_1) = x_1.$$

Their true degree-one Hodge operators are

$$\Delta_0 = \operatorname{diag}(1, 0), \qquad \Delta_1 = \operatorname{diag}(0, 1).$$

Coordinate exchange is a degreewise chain isometry. Both operators have positive spectral gap 1. Standard integral stalk lattices give harmonic lattices $\mathbb{Z}e_1$ and $\mathbb{Z}e_0$, each with an explicit full-span presentation.

Interpret these actual complexes in the bounded-below real cochain Quillen model category. For this example the closed tests ask whether cohomology in an integer degree is nonzero. Their invariance is proved under every typed Helfer equivalence, using quasi-isomorphisms. The chain isometry therefore makes the two actual closed theories equal. This proves invariance for this homological test fragment; it does not formalize Helfer's full first-order formula syntax or general fibrational invariance theorem.

Proposition 3.5 (One observation, two nonzero integral harmonic targets)**.** *Let p be the common closed theory of these two presentations in the stated fragment, and let $b = e_0 + e_1$. There are structured Hodge samples with that same observation (p, b), harmonic messages e_1 and e_0, and zero exact components. Their operative targets are distinct and nonzero. For each presentation's explicit harmonic lattice, its message is also the unique closest lattice vector to b.*

Proof. In the first presentation use harmonic message e_1, exact opening zero and coexact opening 1, producing noise e_0. In the second use message e_0, exact opening zero and coexact opening 1, producing noise e_1. Both sums equal b, and both messages belong to their standard integral harmonic lattices. The actual Hodge projections recover those messages. Orthogonality gives the Pythagorean identity from Proposition 3.2; any other lattice point has strictly positive squared distance from the recovered message. Thus each message is the unique CVP answer in its presented lattice. □

For the same realization, the fixed coordinate e_1 is harmonic for Δ_0 and not for Δ_1. This constructs the separation witness used in Proposition 3.4. The existing no-uniform-recovery theorem therefore applies, and its explicit-CVP-oracle corollary applies under its oracle-existence premise. Equality of the public theory is derived from the actual semantic equivalence, rather than imposed by a one-point classifier. The matrix-level public projection is extended from the two admitted operators; no semantic realization is asserted for other matrices.

The displayed pair uses coefficient 1 in each presentation's own harmonic-lattice basis. Its distinct ambient targets therefore do not alone assert distinct encoded integer plaintexts. A second constructed pair makes the encoding explicit: take $b = e_0 + 2e_1$, harmonic messages $2e_1$ and e_0, and complementary coexact noise e_0 and $2e_1$. Their integer coefficients in the same declared lattice presentations are 2 and 1. Both messages are nonzero, both exact components remain zero, and the public observation is again shared. No function of that observation can return the correct coefficient for both presentations. This is a finite exact-recovery obstruction for the specified integer encoding, not a distributional security experiment. Each message remains the unique CVP answer in its actual lattice.

This example fixes the mathematical observation to (p, b). Its equal zero exact components do not assert equality of complete storage records. The application adapter requires the actual operator and snapshot classifier to agree with the semantic realization before deriving message/operative-target and CVP agreement. Existing payload acceptance, decoder, hash and envelope bindings remain explicit. Paper 2 owns those concrete visibility/storage, rotation and key-delivery profiles; this example establishes no distributional anonymity or plaintext-security experiment.

4 Semantic Classifiers and the Homological Model

4.1 Fibrational Semantics

The path-object and fibrational part of this semantic layer is adapted from Helfer's first-order homotopical logic [Hel25, Sections 2–7]. In particular, we use its replacement of diagonal equality by a path-space fibration, its Grothendieck-fibrational formulation of first-order semantics, and its homotopy-invariance theorem. We do not reproduce Helfer's full development of suitable model categories, Kan complexes, $h^{=}$-fibrations, or two-dimensional fibrations. Instead, we extract the semantic obligations consumed by HCP and add the project-specific public-classifier and observation

boundary below. Helfer's external invariance theorem is cited as background. We retain the typed data needed to apply that result: a multi-sorted algebraic signature, finite-product interpretations, homotopy homomorphisms coherent with every function symbol, and componentwise homotopy equivalence between cofibrant-fibrant interpretations. The closed-formula consequence of Helfer's theorem remains an explicit hypothesis, from which the HCP classifier bridge is proved in Proposition 2.12. We use the usual fibrational notation

$$p : E \longrightarrow B, \qquad E_b = p^{-1}(b)$$

for a total space of hidden data over a base of public or contextual data. When $x, y \in E_b$, equality in the semantic layer is interpreted through a path object rather than by strict equality in the ambient presentation. We write $\mathcal{P}_X(x, y)$ for the path object from x to y in a space or type X, and set

$$x \simeq_b y \quad \Longleftrightarrow \quad \mathcal{P}_{E_b}(x, y) \neq \varnothing.$$

For a dependent family $F : E \to \mathsf{Type}$ equipped with the stated fibrational transport data, a path $\gamma \in \mathcal{P}_E(x, y)$ supplies the chosen map

$$\gamma_* : F_x \longrightarrow F_y.$$

Such transport is additional structure; equality of public classifier values alone supplies no transport map. The protocol realization in Paper 2 uses additional typed transport and observation data; no byte-level construction is inferred here.

The typed interpretation assigns an object of a model category to each sort of an algebraic signature. Finite lists of sorts are interpreted by finite products, and function symbols by maps between those products. A homotopy homomorphism respects those operations up to the specified homotopies; the homotopy-equivalence data used by the classifier bridge also requires the stated componentwise homotopy-equivalence data. These conditions are separate from a relabelling of a presentation or an equality of matrix nullities.

4.2 Path-Object Semantics for Equality

Following Helfer's homotopical interpretation of intuitionistic first-order logic [Hel25, Section 2 and Definitions 6.3 and 6.6], let A be a Kan complex, or more generally an object in a suitable model category, and choose a path-object factorization

$$\Delta_A : A \xrightarrow{\sim} PA \twoheadrightarrow A \times A.$$

In the simplicial realization one may take $PA = A^I$, with the second map given by endpoint evaluation. For points $x, y \in A$, write

$$\mathcal{P}_A(x, y) = PA \times_{A \times A} \{(x, y)\}.$$

The interpretation of the atomic equality formula $x = y$ is this pullback. Its semantic inhabitation condition is therefore

$$x \simeq_A y \quad \Longleftrightarrow \quad \mathcal{P}_A(x, y) \neq \varnothing.$$

As an elementary local consequence, if the chosen path data also carry constant paths, reversal, and composition, then $\simeq_A$ is reflexive, symmetric, and transitive. This equivalence-relation observation is not being quoted as a separate theorem of Helfer.

Helfer's homotopy-invariance theorem applies under its stated categorical hypotheses. Let $M, N : \sigma \to \mathcal{C}_{\mathrm{f}}$ be base interpretations of an algebraic signature in a suitable model category, let $\widehat{M}, \widehat{N}$ be homotopical interpretations over them, and let $M \simeq N$ be a homotopy equivalence of the base interpretations. Then every formula-in-context has homotopy-equivalent interpretation objects over the induced context equivalence. In particular,

$$M \simeq N \quad \Longrightarrow \quad (M \models \varphi \Longleftrightarrow N \models \varphi)$$

for every closed first-order formula φ [Hel25, Definition 7.1 and Theorem 12.3]. We use this external theorem as semantic background; the general HCP comparison retains its closed-formula consequence as an explicit premise.

HCP uses this path-equality semantics as motivation for its hidden-representative model. The public classifier, the cryptographic hiding of a representative or path witness, and the verifier's evidence boundary are project-specific additions; they do not follow from Helfer's invariance theorem alone.

4.3 Public Classifier Fibers

Let A be a type of hidden representatives and let P be a type of public class data. A public classifier is a map

$$\pi : A \longrightarrow P.$$

It determines public fibers

$$A_\alpha = \pi^{-1}(\alpha) = \{x \in A : \pi(x) = \alpha\}, \qquad \alpha \in \mathrm{im}(\pi),$$

and a public equivalence relation

$$x \equiv_\pi y \quad \Longleftrightarrow \quad \pi(x) = \pi(y).$$

The classifier is path-respecting when

$$x \simeq_A y \Longrightarrow x \equiv_\pi y,$$

and nonfaithful when some public fiber contains distinct hidden representatives. A path $\gamma \in \mathcal{P}_A(x, y)$ is a hidden witness that x and y lie in the same path component; under path-respect, its public consequence is the equality $\pi(x) = \pi(y)$. At this stage the public datum is only the classifier value. The protocol profile and the quantitative code-side analysis are developed separately in Papers 2 and 3; this paper retains their observation boundary in Section 6.

Path-respecting gives only the displayed forward implication. Unless a separate completeness condition is imposed, equality of classifier values need not imply the existence of a path; a public fiber A_α may be a coarser union of path components. Accordingly, α denotes the public classifier value and A_α its fiber, not a hidden path witness.

Nor do path-respect and nonfaithfulness prove that π hides x: nonfaithfulness supplies a collision, but it does not imply one-wayness, indistinguishability, or the absence of other private information from the output. Such privacy is a separate cryptographic property of the chosen public-view construction and its distributions.

The fibration projection and public classifier describe compatible but distinct levels. When the hidden representatives are the objects of E, a map $\kappa : B \to P$ on objects supplies the compatibility condition

$$\pi = \kappa \circ p.$$

Then $p(x)$ is the fibration-base point of x, while $\alpha = \pi(x) = \kappa(p(x))$ is its public class label. The special case $B = P$ and $\kappa = \mathrm{id}_B$ identifies the underlying object maps π and p. In general, κ may be lossy, and

$$A_\alpha = \bigcup_{\substack{b \in B \\ \kappa(b) = \alpha}} E_b.$$

This compatibility datum is additional structure; it is not implied merely by the requirement that π respect paths.

4.4 The Constructed Homological Fragment

The concrete family in Proposition 3.5 interprets its single complex sort in the bounded-below cochain model of real vector spaces. Its degree-zero-through-two terms use the actual incidence-defined cellular coboundaries. The closed tests of the declared fragment are indexed by integers: the test in degree j asks whether H^j is nonzero. Quasi-isomorphisms preserve these tests, and the model's weak equivalences are quasi-isomorphisms. The development proves the required cofibrancy and fibrancy for these real complexes and obtains the corresponding closed-test invariance.

Consequently, the degreewise chain isometry exchanging the two coordinates of the constructed example gives equality of the two actual homological theories. This proves the shared-class premise used in the ambiguity argument. It does not identify equal public classes with all possible paths, or establish completeness of the classifier for every hidden representative. The full first-order syntax, equality objects and fibrational invariance statement of Helfer's general framework remain external background; the concrete example proves the named fragment rather than assuming the full first-order theorem.

4.5 Scope of the Semantic Model

The generic bridge in Proposition 2.12 retains its explicit invariance and separation hypotheses. The constructed fragment discharges its own invariance premise; it does not supply protocol authentication, a computational sampler or a security game.

Protocol-specific commitment assumptions, oracle judgments, admissibility gates and proof-carrying state belong to Paper 2 [vdV26b]. Coding predicates, radii and quantitative projection comparisons belong to Paper 3 [vdV26a]. Those developments may reuse these semantic and Hodge objects without becoming additional claims of this manuscript.

5 Observation Boundaries

The recovery statements concern the public pair (α, b) from Definition 2.7. A protocol may instead expose a residue payload, a digest or a proof about a cochain. Those are different observation maps. This section records only the mathematical distinctions needed by the HCP argument. The construction, encoding, visibility policy and security contracts of the cochain commitment belong to Paper 2 [vdV26b].

5.1 Exact Cochains and Full Hodge Samples

Fix one cellular Hodge system and an opening $a \in C^0$. Its exact one-cochain $c = \delta_0 a$ satisfies

$$P_H c = P_H \delta_0 a = 0.$$

Indeed, $\operatorname{im} \delta_0$ is orthogonal to the harmonic subspace. Thus this exact cochain is not the full sample $b = h + e$ when the harmonic message h is nonzero. An encoding modulo an integer, a digest of that encoding and a proof about its acceptance do not, by their types alone, supply the full b or its relation to the operative hidden operator.

There is also a distinction between the exact one-cochain and its opening. Linearity gives

$$\delta_0 a = \delta_0 a' \quad \Longleftrightarrow \quad a - a' \in \ker \delta_0.$$

Consequently the raw coboundary determines an opening only modulo $\ker \delta_0$. A unique degree-zero message encoding requires further application data. This elementary observation is independent of binding a specified exact C^1 payload cryptographically.

To connect a protocol artifact to the HCP observation, an application must supply a relation binding that artifact to the full sample, its operative operator and its classifier value. The artifact can retain an exact component and separately relate it to b; projecting such a related record is not an algorithm that extracts the full sample from residue bytes. Paper 2 gives the concrete payload, snapshot and observation contracts. The argument here uses only the resulting mathematical pair.

5.2 The Exact-Component Obstruction

The loss of information can be quantitative. Under the hypothesis of Proposition 2.6, a nonzero coexact direction gives samples with the same exact component and arbitrarily large full Hodge residual. Hence an observation depending only on that component need not preserve the full-sample residual or a positive far-radius condition. The proposition concerns the stated unrestricted Hodge-noise carrier; a restricted application profile requires its own comparison theorem.

This does not identify the distinct scores

$$\|P_{H^\perp} b\|, \qquad \operatorname{dist}(b, \operatorname{im} \delta_0), \qquad \text{a modular cochain or code distance.}$$

The first measures disagreement with the harmonic subspace, while the second measures distance to exactness. The modular distances and their native

normalizations are specified by the protocol profile and analyzed in Paper 3. No equality of those scores is assumed by the HCP/CVP comparison.

5.3 Additional Public Observations

The constructed ambiguity examples fix (α, b). If a larger released view $V(x, b)$ is used, the same obstruction applies only after proving that the two presentations also give equal values of that view. For example, a view factoring through (α, b) preserves their equality; arbitrary snapshot roots, full geometric coordinates or proof transcripts need not do so. No security conclusion for an unspecified larger record follows.

The full projective point of the operative harmonic subspace is a concrete example of consequential additional information. At the fixed ambient inner product, it determines that subspace's projector and therefore the HCP target from b. We prove this reconstruction next. A digest-only view remains a different observation and is not covered by the reconstruction theorem.

6 Projective Observations and Recovery

The hidden-operator comparison depends on the data actually observed. Full Plücker data expose a subspace, even when no particular basis or matrix is announced. This geometric recovery belongs to the HCP observation argument; the commitment encoding and coding-theoretic consequences are developed in Papers 2 and 3.

6.1 Frames and Projective Coordinates

Let $V = \mathbb{R}^n$ have its fixed Euclidean inner product, and let $w_1, \ldots, w_r$ be an independent frame spanning $W \leq V$. Its exterior coordinate is

$$c = w_1 \wedge \cdots \wedge w_r \in \bigwedge^r V.$$

An independent frame has nonzero exterior product. A different basis of the same subspace changes c by the nonzero determinant of its change-of-basis matrix. The projective class therefore depends only on W:

$$c \sim c' \quad \Longleftrightarrow \quad \exists u \in \mathbb{R}^\times,\ c' = uc.$$

The formal raw carrier is the scalar-orbit quotient of the whole exterior power, including its zero orbit. A genuine frame determines a nonzero, decomposable

orbit. The conclusions below use such frames; they do not assert that every point of the raw quotient is a Grassmannian point.

A related specialized reconstruction uses restricted Plücker coordinates for maximal isotropic subspaces [DK26]. The argument here uses the full projective point of an arbitrary real subspace and the fixed ambient metric; no orthogonal-matroid realization is assumed.

Proposition 6.1 (Projective Plücker invariance)**.** *Two independent frames spanning the same finite-dimensional subspace have equal projective exterior coordinates.*

Proof. Both exterior products are nonzero elements of the one-dimensional top exterior power of that subspace. They differ by a nonzero scalar and hence have the same orbit. □

6.2 Recovery from the Full Point

Proposition 6.2 (Recovery from the full projective point)**.** *Let $w_1, \dots, w_r$ be an independent frame for $W \subseteq \mathbb{R}^n$, and put $c = w_1 \wedge \cdots \wedge w_r$. Then*

$$W = \{x \in \mathbb{R}^n : x \wedge c = 0\}.$$

This reconstruction depends only on $[c]$. Consequently, two genuine rank-r frame witnesses have the same full projective point if and only if they present the same subspace. At the fixed ambient inner product, the full point determines P_W and $I - P_W$.

Proof. If $x \in W$, the frame with x adjoined is dependent and its exterior product vanishes. If $x \notin W$, the adjoined frame is independent and its exterior product is nonzero. Multiplying c by a nonzero scalar does not change this annihilator. The projector is the unique orthogonal projection onto the reconstructed subspace. □

In the raw quotient the zero orbit has the whole ambient space as its annihilator. The frame-indexed statement avoids treating that orbit as an arbitrary nonzero rank-r Plücker point. If $W = \ker \Delta_x$ is the operative harmonic subspace, full projective data together with b determine

$$h = P_W b.$$

The operative-kernel identification and the fixed inner product are essential. A coordinate for an unrelated subspace gives no such conclusion. The full point does not determine the positive spectrum of the original Δ_x. Nor does this exact-real reconstruction give an efficient finite byte algorithm or inversion of a digest.

6.3 Boundary with Protocol and Proximity Analysis

A concrete protocol may map an original cochain payload to a finite received word. Paper 2 owns that payload, its observation/encoding contract and the stage evidence [vdV26b]. Paper 3 owns the quantitative source-to-code comparison, list bounds and MCA radii [vdV26a]. A comparison for its declared modular source profile does not identify an arbitrary Euclidean or full-HCP residual with code distance. In particular, the exact-component obstruction of Proposition 2.6 must still be respected.

The present paper asserts no capacity-scale HCP-to-RS reduction merely from those maps. Any such application needs a stated raw source carrier, its valid locus, its metric and an independently justified far-preservation comparison. The protocol and coding papers retain those hypotheses and quantifier orders in their own statements; the detailed gate, serialization, canonical-field and radius developments are not repeated here.

7 Conclusion and Further Work

The hidden Hodge operator determines the recovery target. An explicitly presented harmonic lattice gives a precise discrete-CVP comparison, while the constructed common-input examples prevent uniform exact recovery from the specified public data. Exact-component observations and full projective observations expose different information, as the two observation results make precise.

7.1 The Three-Paper Boundary

Paper 2 [vdV26b] develops the concrete cochain commitment, observation and encoding profile, verification evidence, recipient rotation and accepted state transitions. Paper 3 [vdV26a] develops list decoding, MCA, radii and quantitative projection comparisons. Those manuscripts reuse the Hodge and semantic objects here without changing the public input of the HCP statements. Protocol privacy must account for its whole released view, and an additional source metric needs its own bridge.

7.2 Complexity and Semantic Refinements

The proved ambiguity is a failure of uniform exact recovery from the specified public data. It supplies neither an asymptotic bit-model hardness reduction nor an average-case encryption-security experiment. Such claims require

encoded instance families, sampling, resource bounds and a precisely stated adversary view. The finite examples and their independent known-lattice CVP answers do not discharge those obligations.

Discrete Morse reduction for cellular sheaf cohomology [CGN15] suggests a computational refinement. A chain-homotopy equivalence alone does not identify the original harmonic representative or preserve its distance. An application to the original recovery target needs explicit retained lifts and a metric comparison.

The constructed classifier proves the named homological-test fragment in the real cochain model. Extending it to the full first-order fibrational semantics, constructing richer families with a declared topological realization, and analyzing transported public observations are separate mathematical tasks. The selected results remain the exact Hodge/CVP comparisons and observation boundaries of this manuscript.

A Statement–Lean Parity Table

Table 2 records the correspondence for this paper's six Hodge/CVP groups and two observation-boundary groups. The independent entry's configuration, `lean/Palomar/PaperOne/comparator.json`, selects 21 manuscript-facing and 25 supporting theorem declarations. Every declaration appears once below. These are coverage counts, not a count of independent research contributions. Definition rows name canonical implementation objects, not additional selected theorem targets. Main-result and support rows are proved under their stated premises; background, assumption and contract rows identify the separate scope boundaries.

Write `E` for `Hcrypt.Palomar.PaperOne`, `CS` for `E.CellularSheaf`, `HB` for `E.HCPBridge`, `AF` for `E.AxisFamily`, and `PL` for `E.Plucker`. The abbreviated anchors expand to the exact configured declaration names.

Table 2: Selected manuscript–Lean parity.

Statement, status and scope	**Corresponding formal declarations or boundary**
Definitions: HCP input and target. Section 2. The public pair contains a classifier value and observed cochain. A private representative selects the operative kernel and hence the orthogonal-projection target.	Canonical definitions: `Hcrypt.HCPBridge.Observation` `Hcrypt.HCPBridge.HiddenHodgeWitness` `Hcrypt.HCP.operativeHarmonicComponent` `Hcrypt.HCPBridge.operativeTarget`
Definitions: explicit-lattice CVP. Section 3.1. An injective real basis presents the lattice through integer coefficients. An instance adds a target; a solution is a closest discrete lattice point.	Canonical definitions: `CVP.ExplicitLattice` `CVP.Instance` `CVP.Solution`
Definitions: semantic realization. Sections 2 and 4. The public classifier factors through the specified closed theory. The operative Hodge operator is separate presentation-level data; equal classifier values need not identify it.	Canonical definitions: `Hcrypt.Axioms.PublicClassifier` `Hcrypt.Helfer.ClosedTheory` `Hcrypt.HCPBridge.Realization`
Definitions: Hodge sample and lattice. Sections 2–3. The sample combines an integral harmonic message with exact/coexact noise. A supplied harmonic-lattice presentation has full real span; it need not contain every intrinsic integral harmonic cochain.	Canonical definitions: `Hcrypt.Sheaf.CellularSheaf.HodgeEncrypte dCochain` `Hcrypt.Sheaf.CellularSheaf.HarmonicLatti cePresentation`

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Table 2: Selected manuscript–Lean parity (continued).

Statement, status and scope	Corresponding formal declarations or boundary
Main result: P1-S01. *Proved.* Proposition 2.5; private recovery in Sections 2–3. The complex equation and actual adjoint Hodge operator give componentwise unique harmonic, exact and coexact parts. Integer coefficient recovery uses the supplied harmonic-lattice presentation.	`E.Hodge_Decomposition_Theorem` `E.Hodge_Decomposition_unique` `CS.HodgeEncryptedCochain.harmonicOneProjection_observed` `CS.HarmonicLatticePresentation.decodeCoefficients_encrypt`
Main result: P1-S02. *Proved.* Proposition 3.1. A field-linear image of the displayed generator is realized as a kernel. The consequence concerns that image, not integer-lattice membership.	`E.exists_matrix_ker_eq_range` `E.Hodge_GeneratedSubspace_Reduction_Proof`
Main result: P1-S03. *Proved.* Proposition 3.2. The discrete lattice is explicitly supplied and lies in the projected subspace. The Pythagorean identity preserves exactly its closest-vector answers.	`E.squaredDistance_projection_decomposition` `E.solution_iff_projectedTarget`
Main result: P1-S04. *Proved.* Cellular specialization of Proposition 3.2. For the actual Hodge sample and supplied harmonic message lattice, the operative message is the unique closest lattice point.	`E.concreteSheafHCP_CVP_solution_iff_message` `E.concreteSheafHCP_CVP_existsUnique`

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Table 2: Selected manuscript–Lean parity (continued).

Statement, status and scope	Corresponding formal declarations or boundary
Main result: P1-S05. *Proved.* Propositions 3.4 and 3.5. Generic separation retains closed-test invariance and kernel separation; oracle corollaries also retain existence of a correct explicit-lattice CVP oracle. The constructed integral family has one public observation and distinct nonzero targets, ruling out uniformly exact public-data recovery.	`HB.operativeTargets_ne_of_kernelSeparation` `HB.cvpOracle_insufficient_of_kernelSeparation` `HB.cvpOracle_insufficient_for_operativeTarget` `AF.cellular_axis_ambiguity` `AF.cvp_oracle_insufficient`
Main result: P1-S06. *Proved.* Weighted example in Section 3.5. The common observation is $b = (1, 2)$, with private integer coefficients two and one. Each privately presented lattice has its own unique CVP answer.	`AF.weighted_axis_recovery_obstruction`
Main result: P1-B01. *Proved.* Proposition 2.6. An allowed nonzero coexact direction prevents the full Hodge residual from factoring through the exact component. The farness obstruction uses the unrestricted noise carrier and exact-component-only extraction.	`CS.HodgeEncryptedCochain.not_residual_norm_factors_exactComponent` `CS.HodgeEncryptedCochain.not_far_preserving_exactComponent`

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Table 2: Selected manuscript–Lean parity (continued).

Statement, status and scope	Corresponding formal declarations or boundary

Main result: P1-B02. *Proved.* Proposition 6.2. A genuine independent frame's full projective coordinates recover its subspace and fixed-metric projector. Operative-target recovery additionally identifies that subspace with the operative kernel. A digest is a different observation.	`PL.recoverSubspace_frame`
	`PL.recoverHarmonicProjector_frame`
	`PL.recover_eq_operativeHarmonicComponent`

Support: complexes and semantics. *Proved.* Sections 2–4. Actual coboundary composition, the constructed complex isomorphism and the corresponding three-term differential.	`CS.delta1_comp_delta0`
	`AF.cellular_complex_iso`
	`E.ThreeTermData.differential_comp`
	`AF.boundarySqZero`

Support: axis coordinates. *Proved.* Section 3.5. Coordinate vectors and the distinct complementary-axis choices in the examples.	`AF.unitCochain_apply`
	`AF.other_zero`
	`AF.other_one`
	`AF.other_ne`

Support: axis Hodge operators. *Proved.* Section 3.5. Both coboundaries, the adjoint, the degree-one Laplacian and its harmonicity criterion.	`AF.delta0_eq_zero`
	`AF.delta1_apply`
	`AF.delta1_adjoint_eq`
	`AF.degreeOneLaplacian_apply`
	`AF.harmonic_iff`

Support: integral lattice. *Proved.* Section 3.5. Restriction-compatible stalk integrality, harmonic basis vectors, encoding injectivity and membership in the chosen lattice.	`AF.standardStalkLattice_contains_iff`
	`AF.restriction_preserves`
	`AF.unitCochain_integral`
	`AF.other_unit_harmonic`
	`AF.harmonicBasis_injective`
	`AF.harmonicLatticeVector_eq`
	`AF.latticeVector_mem`

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Table 2: Selected manuscript–Lean parity (continued).

Statement, status and scope	**Corresponding formal declarations or boundary**
Support: projective quotient. *Proved.* Section 6. Scalar-orbit equivalence laws, wedge-annihilator recovery and invariance under a nonzero scalar multiple.	`PL.ProjectivelyEquivalent.refl` `PL.ProjectivelyEquivalent.symm` `PL.ProjectivelyEquivalent.trans` `PL.mem_wedgeAnnihilator_iff` `PL.wedgeAnnihilator_eq_of_projectivelyEquivalent`
External theorem and semantic premise. Section 4; distinct from the proved homological fragment.	Helfer's general fibrational/path-invariance theorem is cited background. Applying the generic bridge requires the declared invariant closed tests; full first-order invariance and a complete classifier for arbitrary hidden presentations are not proved by this entry.
Assumption: computational hardness. Section 3.1; distinct from the proved exact comparisons.	Known-lattice CVP hardness is an external computational interface. The selected algebraic comparisons and non-identifiability examples do not establish an asymptotic encoded-instance hardness reduction or a sampled encryption-security theorem.
Contract: protocol application. Sections 5–6; distinct from the observation theorems.	A larger released view requires its own equality or factorization proof. Artifact, encoding and visibility contracts belong to Paper 2; quantitative source-to-code comparisons belong to Paper 3. No general capacity-scale full-HCP comparison follows from an exact-component extraction.

B Formal Objects and Proof Boundaries

Independent statements and canonical proofs. `Palomar.PaperOne.Challenge` gives ordinary transparent definitions using Mathlib-only imports. Its finite-product helpers `Hcrypt.Helfer.contextObject` and `Hcrypt.Helfer.contextMap` retain their canonical names. `Palomar.PaperOne.Solution` identifies the independent presentations with the canonical proof objects without importing Challenge. The 23 deliberate Challenge theorem placeholders are comparison targets; Solution has no proof or definition holes.

Source objects and mathematical scope. The finite cellular Hodge operators and integral message presentation are in the `Sheaf` modules. `CVP.lean`, `HCP.lean` and `Hardness.lean` supply the lattice and operative-target comparisons. `Sheaf/HCPAxisFamily.lean`, `HCP/AxisFamily.lean` and `Helfer/CochainModel.lean` construct the actual integral examples and homological-test interpretation. They use the algebraic finite signed-incidence model; no regular CW or simplicial realization of its base is asserted. Integer coefficient recovery is classical and noncomputable. The proved semantic fragment does not supply the general first-order theorem or arbitrary-presentation classifier.

The full-point reconstruction is owned by `Proximity/Concrete/GrassmannianPlucker/Reconstruction.lean`; `TCB/PluckerObservation.lean` identifies the operative target under explicit kernel agreement. The raw projective quotient contains the zero orbit. Recovery uses genuine independent frames, full exact-real coordinates and the fixed ambient metric; it supplies neither digest inversion nor an efficient finite-byte decoder.

Validation applies to exact snapshots. The permitted proof axioms are `propext`, `Quot.sound` and `Classical.choice`; the selected dependency audit reports only these axioms or subsets. The earlier public snapshot `66e1ad2` passed protected preflight. That result does not certify this revised manuscript; its eventual release requires its own exact-commit comparison and report. Mechanical checking is separate from editorial review, novelty assessment and registry registration. The release manifest records the proof/import closure and hashes; the parity table identifies the selected results, rather than every declaration in those supporting modules.

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